Case Study : Smart Financial Market Analytics and Stock Prediction System using Apache Spark

A financial services organization collects large volumes of data from stock exchanges, trading platforms, financial news portals, social media feeds, and historical market databases. The organization wants to develop a **Financial Market Analytics and Stock Prediction Platform** using Apache Spark to analyze market trends, detect anomalies, perform sentiment analysis, and predict stock prices using machine learning and deep learning techniques.

The platform should support large-scale data processing, exploratory data analysis, ETL workflows, database integration, predictive modeling, and cloud-native deployment using DevOps practices.

As a Big Data Engineer, design, implement, and deploy the solution using Apache Spark and PySpark.

Q1. Spark Initialization and Data Loading (2 Marks)

Initialize **Spark Session** and **Spark Context**. Load stock market, financial news, and historical trading datasets and display the schema and sample records.

Q2. RDD Implementation (3 Marks)

Create **RDDs** from financial datasets and perform transformation and action operations to analyze stock transactions, trading volumes, and market movements.

Q3. Key-Value Operations and Persistence (2 Marks)

Implement **key-value pair operations**, perform **shuffle operations**, and demonstrate **RDD persistence** techniques for distributed financial data processing.

Q4. Spark DataFrame Operations (3 Marks)

Convert financial datasets into **Spark DataFrames** and perform selection, filtering, grouping, joins, window functions, and aggregation operations.

Q5. Exploratory Data Analysis and Spark SQL (5 Marks)

Perform EDA and execute Spark SQL queries to:

- Identify top-performing stocks based on market returns.
- Analyze daily and monthly trading volumes.
- Determine sector-wise market capitalization.
- Identify unusual trading activities and anomalies.
- Generate stock price trend reports and moving averages.

Q6. ETL Pipeline Development (3 Marks)

Design and implement an ETL pipeline to:

- Extract stock market data, news feeds, and historical trading records.
- Transform and clean the data.
- Perform feature engineering for predictive analytics.
- Load the processed data into a financial analytics warehouse.

Q7. Machine Learning/Deep Learning Implementation (2 Marks)

Implement a Machine Learning/Deep Learning model to:

- Predict stock prices.
- Forecast market trends.
- Perform financial sentiment analysis.
- Evaluate the model using suitable performance metrics.

Additional Implementation Instructions

A. Version Control using Git and GitHub

- Create a **GitHub repository** for the project.
- Upload all source code, datasets (or dataset links), notebooks, and documentation.

- Maintain proper folder structure and commit history.
- Include a **README.md** file containing project description, installation steps, execution procedure, and output screenshots.

B. Containerization using Docker

- Create a **Dockerfile** for the Spark application.
- Build and test the Docker image.

C. Deployment using Kubernetes

- Create Kubernetes deployment and service YAML files.
- Deploy the application on a Kubernetes cluster (Minikube/Kubernetes/OpenShift).
- Verify successful deployment.

D. DevOps Pipeline Implementation

- Configure a basic CI/CD pipeline using GitHub Actions/Jenkins/GitLab CI.
- Automate build, testing, Docker image creation, and deployment.

E. Project Submission Requirements

Submit:

- Source code
- GitHub repository link
- Dockerfile
- Kubernetes YAML files
- CI/CD pipeline configuration
- Execution screenshots
- EDA and Spark SQL outputs
- Model evaluation report
- Final project documentation